Introduction to Linear, Time-Invariant Systems

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Key Concepts

1) A system maps an input signal to an output signal.

    a) Systems can model physical phenomenon experienced by signals during propagation through the environment, measurement, etc.

    b) Systems can be used to implement a desired effect in the context of a signal-processing algorithm.

    c) The relationship between the input and output signals of a system is normally expressed as a function.

2) A linear system satisfies the principles of proportionality and additivity.

    a) If the input is scaled by a factor a, then the output scales by the same factor a.

    b) The output due to a sum of input signals is equal to the sum of the corresponding individual output signals.

    c) Consequently, the output due to a weighted sum of input signals is equal to the weighted sum of the corresponding individual output signals.

3) A time-invariant system has the same input output characteristic

at all times. If the input is shifted in time, the output experiences the same time shift.

4) The output of a causal system depends only on past/present values of the input.

5) The output of a non causal system depends on future values of the input.

6) There are four common ways to describe the relationship between the input and the output of a system.

a) The difference equation description for discrete-time systems describes the present output as a weighted combination of past outputs and present/past inputs. This form is extremely useful for implementing systems in computation.

b) The impulse response description is based on the output of the system in response to a unit strength impulse input. It is used for both implementing systems and developing intuition.

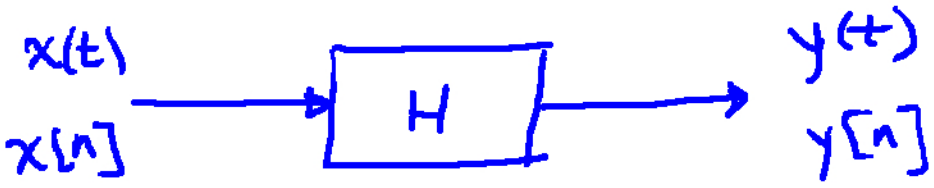
c) The frequency response is based on the complex gain of the system to a complex sinusoid input as a function of the input frequency. It provides a very intuitive description of system characteristics.

d) The system function is based on the complex gain of the system to a complex exponential input $x[n] = z^n$ as a function of the complex number z. It too provides an intuitive description of system characteristics.

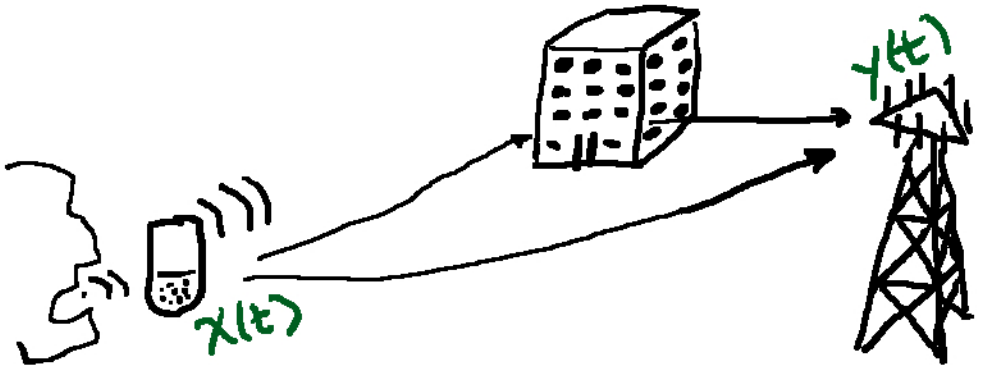
Video Screenshots

0:09

System: Maps an input signal to an output signal



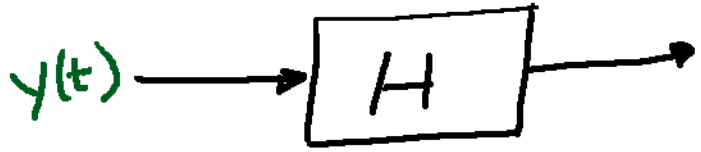
1) Model physical phenomenon



$$y(t) \approx x(t) + \alpha x(t-\tau)$$

$$|\alpha| < 1$$

2) Implement a desired effect



$$z(t) = y(t) - \alpha y(t-\tau) + \alpha^2 y(t-2\tau) - \alpha^3 y(t-3\tau)$$

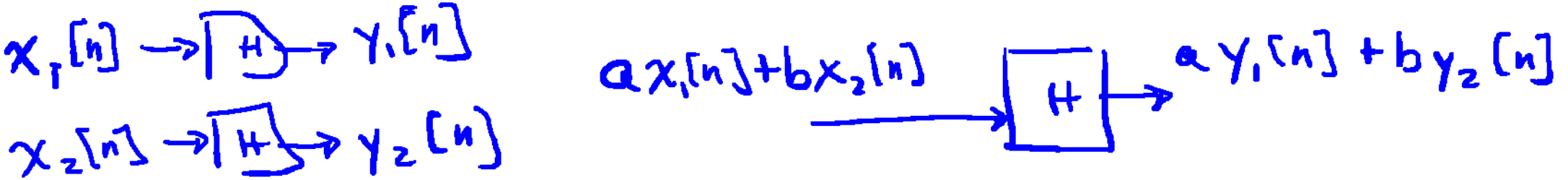
$$\Downarrow$$

$$z(t) = x(t) - \alpha^4 x(t-4\tau)$$

$$\approx x(t)$$

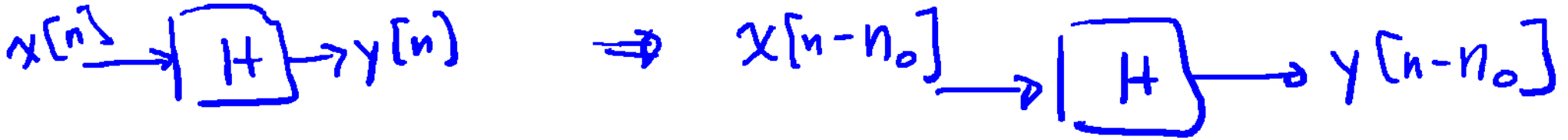
Linear System

Superposition holds: sum of inputs ⇒ sum of outputs



Time Invariant

system responds the same now as it does later



Causal System

- Output depends on past/present inputs

Noncausal: present output depends on future inputs

a) $y[n] = \frac{1}{2}(x[n] + x[n-1])$ Causal

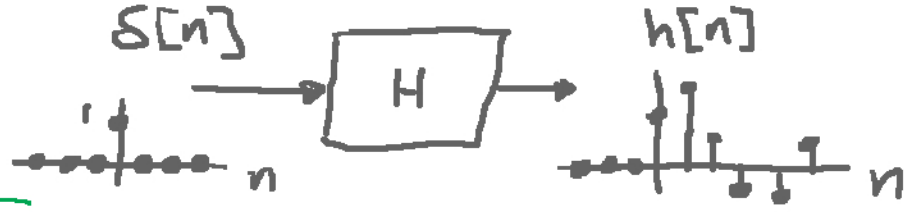
b) $y[n] = \frac{1}{2}(x[n+1] + x[n])$ noncausal

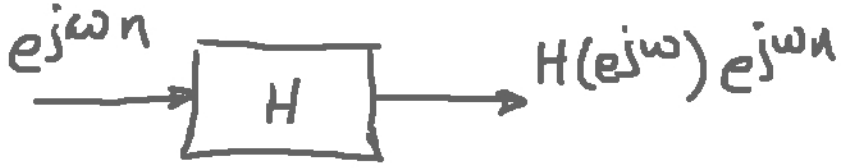
Stored data

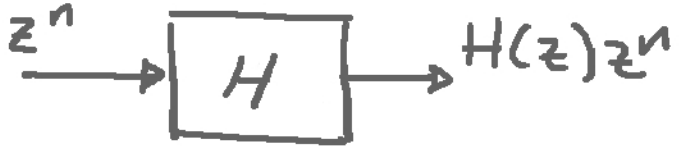
Four LTI System Descriptions

	Computation	Intuition
Difference Equation	★★★★★	★
Impulse Response	★★★	★★★
Frequency Response	★	★★★★★
System Function (poles/zeros)	★	★★★★

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{\ell=0}^{M} b_\ell x[n-\ell]$$







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